

# Comparison of efficacy of different treatment methods in the treatment of idiopathic tinnitus

## İdiyopatik tinnitus tedavisinde farklı tedavi yöntemlerinin etkinliğinin karşılaştırılması

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**Objectives:** This study aims to detect whether any differences were present between betahistine dihydrochloride, transcutaneous electrical nerve stimulation and pure tone masking-tinnitus retraining therapy (TRT) methods in the effects on quality of life and treatment of the symptoms of the patients.

**Patients and Methods:** A total of 91 patients (42 females, 49 males; mean age 49.3±8.3 years; range 30 to 70 years) who admitted to the Otorhinolaryngology Clinic of the Ufuk University between June 2009 and June 2010 with a complaint of subjective tinnitus and who had no hearing loss were included in the study. In this study, the effects of these three treatment methods on healing and quality of life in patients suffering from bilateral subjective tinnitus were comparatively evaluated using Tinnitus Handicap Inventory Score (THIS), visual analog scale (VAS) and audiological parameters. The evaluations were made immediately before the treatment, immediately after the treatment and three weeks after the treatment. Kolmogorov-Smirnov analysis was used to test the normal distribution of the data and Wilcoxon signed rank test was used to show the differences between the different treatment methods before the treatment, immediately after the treatment and three weeks after the treatment. Mann-Whitney U and Kruskal-Wallis H tests were used to show the inter-group differences.

**Results:** In the inter-group analyzes, success rate of the pure tone masking-TRT was much higher when compared to the other treatment methods. In the evaluations performed at the end of the three-month period, it was seen that the efficacy of the treatment was continuing.

**Conclusion:** According to these results, pure tone masking-TRT was found to be the best treatment method when compared to other methods and it was concluded that this treatment may be considered as the first choice in patients with idiopathic tinnitus.

**Key Words:** Betahistine dihydrochloride; pure tone masking; quality of life; tinnitus retraining therapy; transcutaneous electrical nerve stimulation.

**Amaç:** Bu çalışmada betahistin dihidroklorid, transkütaneal elektriksel sinir stimülasyonu ve saf ses maskeleyme-tinnitus eğitim terapisi (TET) yöntemleri arasında hastaların yaşam kalitesi ve septomların tedavisi üzerindeki etkiler açısından fark olup olmadığı belirlendi.

**Hastalar ve Yöntemler:** Haziran 2009 - Haziran 2010 tarihleri arasında Ufuk Üniversitesi Kulak Burun Boğaz Kliniğine subjektif tinnitus yakınması ile başvuran ve işitme kaybı olmayan toplam 91 hasta (42 kadın, 49 erkek; ort. yaş 49.3±8.3 yıl; dağılım 30-70 yıl) çalışmaya dahil edildi. Bu çalışmada iki taraflı subjektif tinnitus yakınması olan hastalarda bu üç tedavi yönteminin iyileşme ve yaşam kalitesi üzerine etkisi Tinnitus Değerlendirme Anketi Skoru (THIS), görsel analog ölçek (GAÖ) ve odyolojik parametreler kullanılarak karşılaştırılmalı olarak değerlendirildi. Değerlendirmeler tedaviden hemen önce, tedaviden hemen sonra ve tedaviden üç hafta sonra gerçekleştirildi. Verilerin normal dağılımını test etmek için Kolmogorov-Smirnov analizi, değişik tedavi yöntemleri arasındaki tedavi öncesi, tedaviden hemen sonra ve tedaviden üç hafta sonraki farkı göstermek için Wilcoxon signed rank testi kullanıldı. Gruplar arası farklılıkları göstermek için Mann-Whitney U ve Kruskal-Wallis H testi kullanıldı.

**Bulgular:** Gruplar arası analizlerde saf ses maskeleyme-TET yönteminin başarı oranı diğer yöntemlerle karşılaştırıldığında çok daha yüksek idi. Üç aylık periyodun sonunda yapılan değerlendirmelerde de tedavi etkinliğinin devam ettiği görüldü.

**Sonuç:** Bu sonuçlara göre saf ses maskeleyme-TET'nin diğer yöntemlerle karşılaştırıldığında en iyi tedavi yöntemi olduğu bulundu ve bu tedavinin idiyopatik tinnituslu hastalarda ilk tedavi seçeneği olarak değerlendirilebileceği sonucuna varıldı.

**Anahtar Sözcükler:** Betahistin dihidroklorür; saf ses maskeleyme; yaşam kalitesi; tinnitus eğitim tedavisi; transkütaneal elektriksel sinir stimülasyonu.

Tinnitus is classically defined as ringing, buzzing, or other sounds without an external cause. Tinnitus, one of the most common symptoms of the auditory system, affects 17% of the entire society and 33% of the old aged society.<sup>[1]</sup>

Tinnitus, also called syrigmus has been described since ancient Egypt, however, it was first defined as a pathologic symptom by an otorhinolaryngologist Wegel in an article published in the Archive of Otolaryngology in 1931.<sup>[1]</sup>

Tinnitus is clinically classified as objective or vibratory (which can be heard by the patient and the doctor examining the patient) tinnitus and subjective and non vibratory (which can be heard only by the patient) tinnitus. Formation mechanisms of these two types of tinnitus are completely different from each other.

In objective tinnitus, the doctor also hears the sound heard by the patient. The sound can sometimes be heard by means of a stethoscope, as its intensity may be low. Eustachian tube and vascular anomalies are among the reasons for dysfunctions of the muscles in the tympanum and palatal muscles.<sup>[2]</sup> Subjective tinnitus is encountered more than objective tinnitus. Otological, neurological, pharmacological and psychological etiologies underlie subjective tinnitus.<sup>[3]</sup>

Researchers have attempted to describe formation and detection of tinnitus by means of various mechanisms. Tinnitus may be examined in two groups as those which result from para-auditory structures (vascular anomalies and neurovascular anomalies) and from sensorineural auditory system (otologic, neurologic, pharmacological, dental and psychological reasons) according to the pathologic physiology of the disease.

Treatment in tinnitus, if etiology could be detected, must be directed to etiology. On the other hand, etiology in tinnitus may be determined in only 5% of the cases.<sup>[4]</sup>

If etiology cannot be detected, various symptomatic treatment methods are used in patients suffering from chronic subjective idiopathic tinnitus.

While there is still no single pharmaceutical agent or method whose efficacy is acknowledged by all authors, there exist various treatments

for tinnitus.<sup>[3]</sup> Among these tinnitus treatment choices are pharmacological treatment (betahistine dihydrochloride, ginkgo biloba extract, local anesthetics, anticonvulsants, antidepressants, sedatives, myorelaxants, vasodilators, vitamins etc), masking treatment, tinnitus retraining therapy (TRT), psychotherapy, electrotherapy, transcutaneous electrical nerve stimulation (TENS), surgical methods, etc.

The extent of tinnitus complaints in society and lack of success of treatment methods in its treatment render otorhinolaryngologists, audiologists and patients desperate. Thus, the current pursuit of different treatment methods continues.

This study comparatively evaluated the therapeutic effects of oral betahistine dihydrochloride, pure sound masking-TRT and transcutaneous electrical stimulation methods in patients suffering from chronic subjective idiopathic tinnitus.

## PATIENTS AND METHODS

Patients who had applied with bilateral tinnitus complaints for at least six months to the Ufuk University Medical Faculty Hospital Otorhinolaryngology clinic between June 2009 and June 2010 were evaluated.

The Institutional Review Board of Ufuk University Medical School approved the present study and all patients signed informed consents before entering the study.

Detailed medical histories of the patients included in the study were taken and their otorhinolaryngology examinations were made to search tinnitus etiology. Those suffering from any systemic disease (hypertension, diabetics, hyperlipidemia, hypothyroidism-hyperthyroidism), those having histories of otorhinolaryngologic pathologies, those possessing objective tinnitus reasons (arteriovenous malformation, glomus tumor, aneurysm, palatal myoclonus, patulous eustachian tube and muscular contraction in tympanum etc.), presbycusis, history of exposure to loud noise, Ménière's disease, otosclerosis, head trauma, acoustic neuroma, ototoxic drugs, barotrauma and those suffering from tympanum effusion, temporomandibular joint disease and known neuropsychiatric disease were excluded from the study.

Pure audiometric examinations were performed on the patients. Two hundred and fifty, 500, 1000, 2000, 4000 and 8000 Hz pure tone hearing thresholds for air conduction and 500, 1000, 2000, 4000 Hz pure tone hearing thresholds for bone conduction were evaluated by means of Interacoustic Clinical Audiometer AC (Interacoustics, Denmark, 2007) 33 device. Pure tone environments, air and bone conduction hearing thresholds, speech discrimination scores and speech reception thresholds were evaluated. Tinnitus frequencies and amplitudes of the patients were detected. Patients whose pure tone hearing thresholds were 20 dB and above were included in the study. The patients with pathologically audiological findings were excluded from the study. Tympanometry and acoustic reflex measurements were performed using Interacoustic Clinical Audiometer AZ 26 (Interacoustic AZ 26, Denmark). The patients in whom a type A tympanogram curve and stapedial reflexes in all frequencies could be obtained were included in the study. After the conclusion of treatments of the patients included in the study, all audiological tests were repeated. Cranial magnetic resonance images were obtained in all of the patients to eliminate possible pathologies in their posterior fossa, temporal lobe and internal acoustic canal areas.

A total of 91 patients (42 females, 49 males; mean age  $49.3 \pm 8.3$  years; range 30 to 70 years) suffering from subjective idiopathic tinnitus, were included in the study. The patients were chosen and randomly divided into three groups based on treatment methods. Thirty-eight of whom 17 were women and 21 were men, constituted the first group to be treated with betahistine dihydrochloride, 37 of whom 16 were women and 21 were men, made up the second group to undergo TRT treatment (with pure sound tinnitus masking treatment) and 16 of whom nine were women and seven were men constituted the third group to receive TENS treatment.

Betahistine dihydrochloride in the form of 24 mg tablets bid perorally were given to the first group for two months. No additional treatment was applied apart from that.

Pure sound masking treatment was applied to the second group eight times at one-week intervals, each treatment lasting for an average of 60 minutes. The conformity of the frequency and intensity

of the tinnitus masking sound applied were determined. Suppression of the tinnitus by applying masking sound was maintained. The sound given for masking was narrow band pure tone with the lowest intensity and in the tinnitus frequency which masked the tinnitus. Total treatment duration was two months. Unlike the other groups, this group was given the opportunity to communicate with the physician before and after each session regarding their examinations and observations and they were reassured that they had no underlying serious disease and their sufferings would terminate. They were reminded that they should remain in completely silent environments for three months and they were ensured that they could communicate with the physician who had treated them whenever they wished until the last evaluation.

Finally to the third group, the TENS treatment was applied by placing a negative electrode on the mastoid and positive electrode over the temporomandibular joint, giving 0-60 milliamp electric current stimulation having 30 Hz frequency. A two channel TENS device was used in the patients since they were suffering from bilateral tinnitus. Each session lasted for 30 minutes and 14 sessions were performed. Total treatment duration was two months.

The tinnitus evaluation survey (THI=Tinnitus handicap inventory),<sup>[5]</sup> validity and reliability of which were acknowledged, was administered to the patients to determine the intensity of tinnitus and adverse effects of tinnitus on the patients. Adverse effects of tinnitus on the patients were evaluated and graded with visual analog scale (VAS).

Tinnitus handicap inventory is a survey consisting of 25 questions and each of it includes the choices of "Yes", "No" and "Sometimes". Total score of the survey is obtained by addition of the choices marked in each question ("Yes" 4 points, "No" 2 points and "Sometimes" 0 point). The lowest score of the survey is 0 and the highest score is 100. The evaluation was as follows: if the score obtained is 0-16, the level is 1 [Very low (only heard in silent environment)]; if 18-36, level is 2 [Low (easily forgettable by environmental sounds)]; if 38-56, the level is 3 [Medium (noticeable by the background noise, however, not preventing performance of daily activities)]; if 58-76, the level is 4 [Severe (almost heard

continuously, resulting disturbance during sleep and preventing performance of daily activities)] and if 78-100, level is 5 [Very severe (heard all the times, disturbing sleep level, resulting in difficulty in any activity)].

To test whether the data was distributed normally, Kolmogorov-Smirnov analysis; to test whether operations differences in the groups were significant, Wilcoxon signed rank test and for inter-group differences, Mann-Whitney U and Kruskal-Wallis H tests were made using SPSS for Windows (SPSS Inc., Chicago, Illinois, USA) 17.0 statistical package.

## RESULTS

Tinnitus frequency averages of the patients included in the study was  $6065.93 \pm 1443.77$  Hz and tinnitus intensity average was  $43.79 \pm 10.75$  dB HL. There was a homogenous distribution among the tinnitus frequency and intensity values of the patients included in the study.

In our study, in the evaluations made with THI upon treatment of 16 patients to whom TENS was applied, decrease in tinnitus was detected in seven patients (43.75%). Immediately following the treatment, VAS scores maintained their high levels in nine patients (56.25%) and VAS scores three months later also maintained their high levels in nine patients (56.25%).

In our study, in the evaluations made with THI upon treatment of 38 patients to whom betahistine dihydrochloride treatment was applied, decrease in tinnitus was detected in 13 patients (34.21%). Immediately following the treatment, VAS scores maintained their high levels in 17 patients (44.73%) and VAS scores three months later maintained their high levels in 11 patients (28.94%; Figure 1).

In our study, in the evaluations made with THI upon treatment of 37 patients to whom TRT was applied, decrease in tinnitus was detected in 36 patients (97.29%). Immediately following the treatment, VAS scores maintained their high levels in 37 patients (100%) and VAS scores three months later maintained their high levels in 35 patients (94.59%).

Tinnitus handicap inventory scores in the first group receiving betahistine dihydrochloride treatment were reduced from 3.44 to 3.06 (difference average:  $0.38 \pm 0.61$ ), THI scores in the second group given masking treatment were

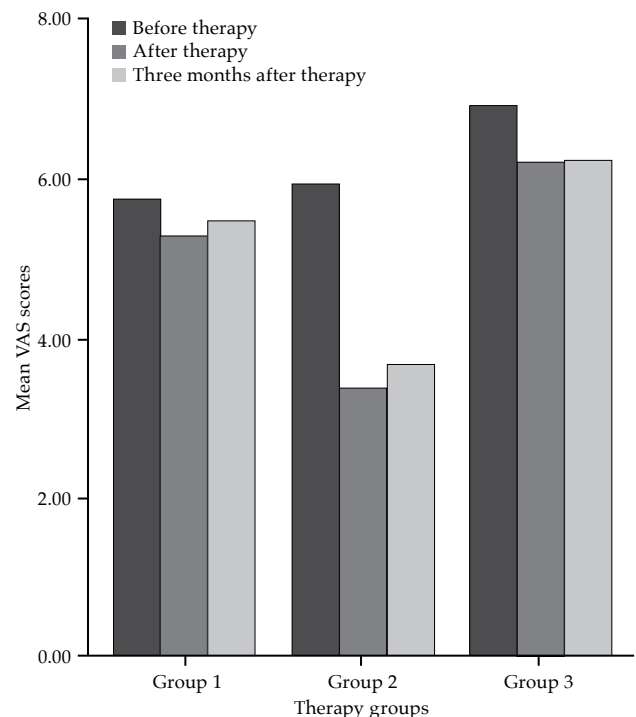
reduced from 3.50 to 1.75 (difference average  $1.75 \pm 0.72$ ) and THI scores in the third group to whom TENS treatment was applied were reduced from 3.55 to 3.11 (difference average:  $0.44 \pm 0.53$ ; Figure 2). The difference among THI scores before and after the treatment was found statistically significant ( $p < 0.05$ ; Figure 3).

When inter-group analysis was made, a significant difference was found based on THI results between the groups receiving betahistine dihydrochloride treatment and the groups receiving TRT + pure sound masking, and a significant difference was also found between the groups receiving sound masking and TENS treatment ( $p < 0.05$ ). No meaningful difference was found between patient groups receiving betahistine dihydrochloride treatment and those receiving TENS treatment ( $p > 0.05$ ).

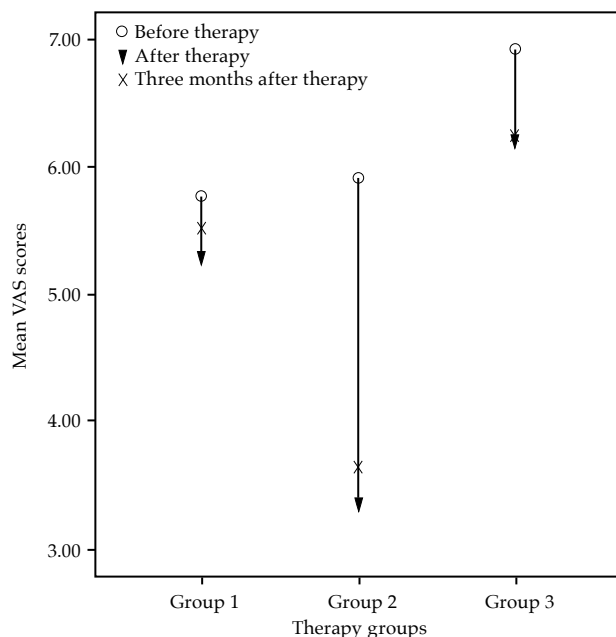
## DISCUSSION

Subjective tinnitus is defined as hearing a sound without existence of an external stimulus.<sup>[6]</sup>

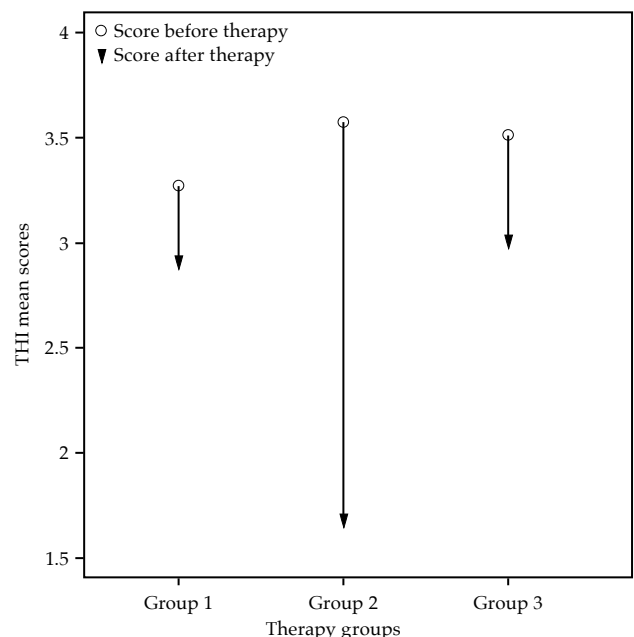
Despite the advances upon Jastreboff's tinnitus definition, tinnitus generation mechanisms have



**Figure 1.** Differentiation of mean VAS scores before, after and 3 months after therapy. VAS: Visual analog scale; Group 1: Patients receiving Betahistine dihydrochloride; Group 2: Patients receiving tinnitus retraining therapy and pure tone masking treatment; Group 3: Patients receiving transcutaneous electrical nerve stimulation.



**Figure 2.** Decrease of mean VAS scores before and after therapy. VAS: Visual analog scale; Group 1: Patients receiving betahistine dihydrochloride; Group 2: Patients receiving tinnitus retraining therapy and pure tone masking treatment; Group 3: Patients receiving transcutaneous electrical nerve stimulation.



**Figure 3.** Decrease of THI scores before and after therapy. THI: Tinnitus handicap inventory; Group 1: Patients receiving betahistine dihydrochloride; Group 2: Patients receiving tinnitus retraining therapy and pure tone masking treatment; Group 3: Patients receiving transcutaneous electrical nerve stimulation.

not been set forth. The difficulty in analyzing source of tinnitus and the reasons triggering it has resulted in an inability of finding effective methods of treatment. This is the principal reason for seeking different treatment methods in the treatment of tinnitus.<sup>[6]</sup>

Since etiology mostly cannot be detected in tinnitus, tinnitus treatment is symptomatic. The prime purpose of treatment is to eliminate the tinnitus. Nevertheless, this is mostly not possible and what is henceforth attempted is trying to maintain tinnitus at a minimum level to minimally affect the life of the patient. Among the pharmaceutical agents, betahistine hydrochloride, ginkgo biloba extract, lidocaine, antilutamatergic agents, dopamine agonist and antagonists, melatonin, zinc preparations and many other agents are used in treatment.<sup>[7]</sup>

Apart from pharmacotherapy, cognitive and behavior treatment,<sup>[8]</sup> sound treatment,<sup>[9]</sup> hearing aids<sup>[10]</sup> and electrical suppression are also used.

Tinnitus is a subjective complaint demonstrating mutual interaction with human psychology. Thus, it is highly difficult to detect tinnitus severity and its effect levels on the life quality of the patient complaining of tinnitus. Before and after treatment,

we evaluated the tinnitus complaints with tinnitus evaluation survey and VAS, validity and reliability of which were acknowledged.<sup>[11]</sup>

Tinnitus is mostly encountered between the ages of 40-80.<sup>[1]</sup> Both genders are equally affected by tinnitus.<sup>[12]</sup> In this study, the average age of the patients included in the study was  $50.6 \pm 8.9$  and women/men ratio was 0.74, which is in accordance with the literature.

Tinnitus may occur together with hearing loss<sup>[13]</sup> and is encountered more in patients suffering from hearing loss.<sup>[14]</sup> It is considered that tinnitus in patients not suffering from hearing loss is based on damage to external hair cells in the cochlea.<sup>[15]</sup> Tinnitus is not accompanied by hearing loss in the patients included in our study and there may be external hair cell damage in these patients.

Tinnitus frequency averages differ in the studies conducted, i.e. while the average of tinnitus frequencies was  $5796.67 \pm 3017.89$  Hz in one study,<sup>[16]</sup> this average was 2718 Hz in another study.<sup>[10]</sup> In our study, this average was calculated at  $6065.93 \pm 1443.77$  Hz. Since tinnitus frequency may differ even in the same patients, tinnitus frequency average was not included among the correction criteria in our study.

Tinnitus is most intense at the hearing threshold of the tinnitus frequency or 20 dB more than that threshold. Tinnitus amplitude average in our study is detected at  $43.79 \pm 10.75$  dB HL. When it is considered that hearing of the patients are in normal ranges, the amplitude average found in our study is in the normal range as well.

Tinnitus is a real perception with neuro-otologic origin. However, the psychological state of the patients are generally affected. Psychological reasons which in turn increase anxiety are involved in the prognosis of tinnitus. On the other hand, the existence of tinnitus may result in frustration, stress, concentration loss, sleep disorder and depression. In other words, there is a mutual interaction between tinnitus and psychological state of a person. In order to break this vicious circle, informing the patient and psychological support given to the patient are of great importance.

Tinnitus retraining therapy is a treatment based on Jastreboff's neurophysiologic tinnitus model. In this treatment method, the disturbing effect of tinnitus is suppressed by exogenous sound energy produced in different forms.<sup>[1]</sup>

It was proved by studies that sounds given externally for treatment purposes increase inner blood flow and decrease tension of neural activity.<sup>[17]</sup>

Masking treatment was an earlier method which was used for the first time in the late 1970's and which became popular in the 1980's.<sup>[18]</sup>

In tinnitus masking treatment, tinnitus is intended to be imperceptible using a sound in the tinnitus frequency and in the same amplitude. The reason for suppression of tinnitus in masking treatment cannot be explained. However, the main purpose in this treatment is to reduce the tension and stress resulting from tinnitus by external sounds.<sup>[19]</sup> Therefore decrease in stress and tension helps both to maintain resistance to tinnitus and to break the triggering mechanism which causes a vicious circle resulting in tinnitus. This explanation is seen to be in conformity with Jasterboff's tinnitus model.<sup>[4]</sup> The effect obtained at the end of the treatment is decrease in the complaints and increase in the life quality of the patient.

Masking treatment is used if the hearing of the patient is in the normal ranges, if not, hearing aids may be used. Different therapeutic sound

sources are recommended to the patient such as CD's, tabletop fountains, sound machines, sound pillows, and so on.

Researchers obtained various results in the researches they applied using different methods in different patient groups. According to the studies made by Vernon et al.,<sup>[19]</sup> in 91% of the patients, masking the tinnitus was possible.<sup>[20]</sup> In a different study, 58% improvement was observed through masking treatment.<sup>[21]</sup>

Counseling offered to the patient along with tinnitus masking (TM), is proven to contribute to patient relief in the studies.<sup>[22]</sup>

In our study, treatment is designed for patients in a line between masking and TRT. For this purpose, aspects which we thought were successful of both treatment methods were used and a treatment protocol was prepared. The principal reason for this was the different results obtained in the studies performed up to now.

There are two<sup>[2]</sup> basic elements of TRT method; training, and consultancy and sound therapy.<sup>[23]</sup>

Contrary to TM, treatment in TRT differs in that the sound given to the patient is not very intense and does not mask tinnitus. However, there is always a sound in the background. The sound in the background reduces the contrast between tinnitus and silent environment. Thus, tinnitus stands out less. This may be defined as a passive attention distraction. The patient reduces the disturbing perception of tinnitus and stress it causes using this background sound for as long as possible everyday. In this study, masking sound used in TRT is proven to suppress the neural activity in connection with tinnitus.<sup>[14]</sup>

The patients to be treated with TRT are divided into five groups: 0, 1, 2, 3 and 4. Counseling is given to all of the patients and depending on the categories to which the patients belong, different treatment methods are used. In the TRT, patients in category 0 complain of low intensity tinnitus and counseling is considered to be sufficient in this group. Patients in category 1 complain of severe tinnitus. They are supported with sound generating hearing aids of low intensity which contain broadband noise in the background. Patients in category 2 suffer from hearing loss along with severe tinnitus. Their treatment takes place with hearing aids to which therapy instruments are added. The basic problem of patients in

the 4<sup>th</sup> category is hyperacusis. Therapy attempts to increase their tolerance levels using special desensitization techniques. Patients in category 3 hear sounds around them in an exaggerated manner as if they were coming from explosives. Low energy sounds are used in this group and attempts are made to decrease their reactions to sound.<sup>[23]</sup>

All of the patients in our study were in categories 0 and 1. Therefore, radio and television were used for the patients constituting the two treatment groups for as long a duration as possible but not less than three days. Besides, patients were ensured to receive counseling and to communicate with physicians whenever desired. Therefore, a mixture of the method used in TRT and that used in TM was formed. This application does not seem possible in patients constituting categories 2, 3 and 4 who suffer from hearing loss, hyperacusia, or sound distortion.

At the end of this treatment, a high success rate was obtained in patients who did not have any additional complaints except for tinnitus in categories 0 and 1. In our study, in the evaluation made using TIHS made upon treatment of 37 patients to whom TRT and pure masking treatment were undertaken, improvement was observed in 35 patients (97.3%). In their evaluations using VAS immediately after treatment, increase in tinnitus was observed in 37 (100%) patients and three months later, in their evaluations made again using VAS, increase in tinnitus was observed in 35 patients (94.59%).

This evaluation is a subjective one made by the patient. In addition, only the patients' feeling better in respect to their previous states was evaluated as a positive response to treatment in our studies.

Follow-up duration for the patients was three months in our study. At the end of the third month the values derived using VAS and those values derived using the same scale at the end of the treatment third month were very close to each other. The success of this treatment method needs to be observed for a longer time and new studies are required in this respect.

Transcutaneous electrical nerve stimulation treatment is a treatment which aims at reduction of tinnitus by electrical stimulus. Sensory fibres are stimulated in this treatment.<sup>[24]</sup> Transcutaneous electrical nerve stimulation method is based on stimulation of dorsal cochlear nucleus through C2

somatic nerve roots and showing of an inhibiting effect of this stimulation on tinnitus. Hence a spontaneous discharge in vestibulocochlear nerve peripheral sensory branch generated by hyperpolarization takes place, thus decreasing tinnitus.<sup>[25]</sup>

Tinnitus treatment using electrical stimulus is a method used for the last 30 years (Portmann et al. 1983). In 1983, Portmann et al.<sup>[26]</sup> reported that he received 66% positive response in his patients suffering from tinnitus to whom he applied electric stimulation from the round window.

In 1985, Engelberg and Bauer<sup>[27]</sup> applied TENS to 20 patients suffering from tinnitus and received positive response from 85% of them. Kuk et al.<sup>[28]</sup> applied alternating electric current to the tympanic membranes of 10 patients and received responses from 50% of them. In other studies, 28%<sup>[24]</sup> and 53%<sup>[28]</sup> decreases were achieved in tinnitus by preauricular and postauricular electrical stimulation.

It is stated that its effect will increase when point of application is closer to the cochlear neurons.<sup>[29]</sup> However negative effects of possible cochlear damage should be noted. Another factor affecting success in TENS treatment is the quality of the current given. Direct current was found more effective compared to alternating current but its cochlear damage potential is higher than alternating current and it is recommended to be used only in patients suffering from tinnitus along with hearing loss.<sup>[30]</sup> We used alternating current in our patients as they did not suffer from hearing loss.

In studies made by different researchers, various results have been obtained using various methods.<sup>[31]</sup> As can be seen from these studies, a standard treatment method in TENS applications have not yet been developed.

In our study, TIHS evaluations of 16 patients upon TENS treatment, improvement was observed in seven patients (43.75%). A continuing decrease in tinnitus was observed in nine patients (65.25%) immediately after the treatment and in nine patients (65.25%) three months later.

Betahistine is an H1 receptor antagonist. Betahistine is used in tinnitus due to vascular effects of dihydrochloride. It dilates precapillary sphincter in the veins and increases inner ear vasculature.<sup>[6]</sup> In a study, betahistine

dihydrochloride treatment was proven to be significantly effective when compared to placebo.<sup>[8]</sup> In another study, a positive response was received from approximately half of the cases with the combined use of betahistine, B complex vitamins<sup>[13]</sup> and diazepam.<sup>[32]</sup>

In our study, in the TIHS evaluations of 38 patients upon TENS treatment, improvement in 13 patients (34.21%) was observed. A continuing decrease in tinnitus was observed in 17 patients (44.73%) immediately after the treatment and in 11 patients (28.94%) three months later.

In our study, betahistine dihydrochloride was observed to be less successful when compared to the other treatment methods.

According to the results of our studies, the most successful treatment method is TRT + pure sound masking combination applied to the second group. It is understood that in almost all of the patients tinnitus complaints decreased, considering the changes in THI and VAS scores. This high success rate could not be observed in other groups.

Another factor affecting success in the second group may be the constant relationship between the patient and physician. The same opportunity was not offered to the first group receiving drug treatment or the third group receiving TENS treatment. Maintenance of morale support in patients in the second group might have resulted in an increase in motivation for healing.

Consequently, in patients suffering from subjective idiopathic tinnitus but not from hearing loss, combined use of TRT and pure sound masking treatment is highly effective and should primarily be preferred. Transcutaneous electrical nerve stimulation treatment and betahistine dihydrochloride treatment should also be taken into account as alternative methods.

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